Financial Time Series
46-929
Final Exam

Wednesday, May 6, 2020, 10:00-11:30AM.

1. Suppose $\{X_t\}$ is a weakly stationary time series. Which of the following is/are definitely true? For each part, assume s and t are integers.

   Choose as many as are appropriate including NONE

   a) ** $E(X_s) = E(X_t)$ for all s and t.
   b) ** $Var(X_s) = Var(X_t)$ for all s and t
   c) $Cov(X_t, X_{t+1}) = 0$ for all t
   d) $Var(X_t|X_{t-1}) = Var(X_s|X_{s-1})$ for all s and t.
   e) ** $Cov(X_t, X_{t+s}) = Cov(X_t, X_{t-s})$ for all s and t
   f) $\{X_t\}$ is a martingale difference sequence (MDS).

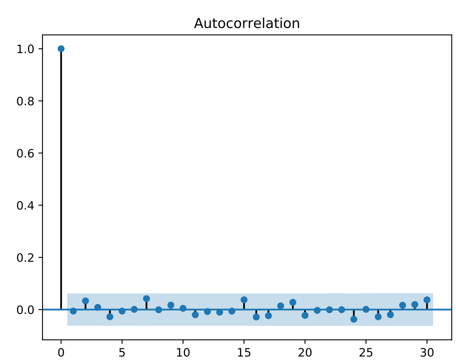
2. Suppose that $\{\epsilon_t\}$ is weak white noise and $\{X_t\}$ is a time series given by

$$X_t = \epsilon_t + \epsilon_{t-1}.$$

   Which of the following is definitely true?
   Choose as many as are appropriate including NONE

   a) ** $\{X_t\}$ is weakly stationary
   b) $\{X_t\}$ is a weak white noise process
   c) $\{X_t\}$ is a martingale difference sequence (MDS)
   d) $\{X_t\}$ is independent white noise

3. Suppose that the vector **X** holds a time series of length 1000. The ACF is given below along with data from the Statsmodels Ljung-Box test applied to this series.



smm.stats.diagnostic.acorr_ljungbox(X),lags = 30)
(array((0.03272623, 1.14287011, 1.21465912, 1.98365807, 2.01716363,
2.0179659 , 3.78124128, 3.78189992, 4.07338676, 4.09649843,
4.50061302, 4.55908731, 4.66279071, 4.6990521 , 6.13059313,
6.96355874, 7.53005575, 7.73329434, 8.5247657 , 9.03487107,
9.04500491, 9.04547543, 9.04569321, 10.45722053, 10.45802865,
11.24878095, 11.64202263, 11.92590159, 12.30728807, 13.73446421,
)), array((0.85644299, 0.56471446, 0.74949064, 0.73876478, 0.84676506,
0.91803888, 0.80460122, 0.87624554, 0.90651627, 0.94288877,
0.95292617, 0.97111117, 0.98188602, 0.98957587, 0.97746958,
0.97395709, 0.97559118, 0.98234778, 0.98056736, 0.98249985,
0.98885553, 0.99309099, 0.99580104, 0.99238751, 0.9952246 ,
0.99467405, 0.99554936, 0.99656001, 0.9971477 , 0.99516411,)))

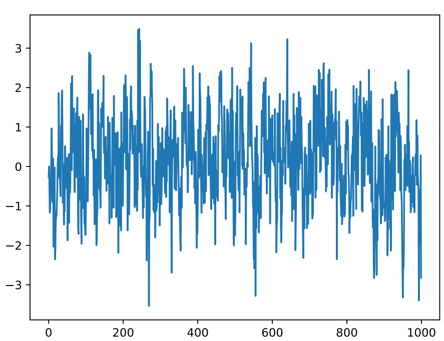
Which of the following is/are true?
Choose as many as are appropriate including NONE

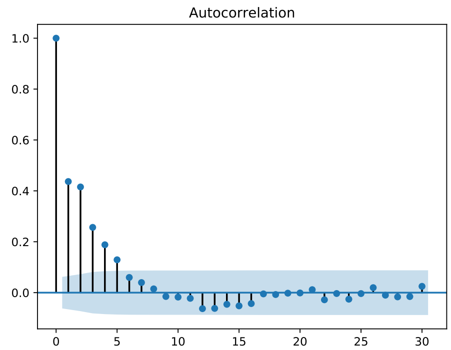
a) There is strong evidence that there is non-zero correlation in the series at some positive lag.

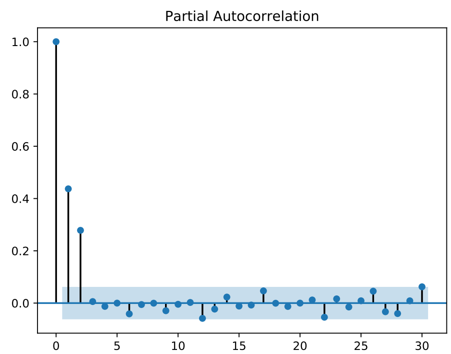
b) ** There is a lack of strong evidence that there is non-zero correlation in the series at some positive lag.

c) These results indicate the clear need for an ARCH/GARCH component to model this series.

d) These results indicate strong evidence that the series is not stationary.







4. The figures above show a time series plot along with the associated ACF and PACF. Using just these plots, which of the following time series models is the best model that fits the data? **Choose exactly one answer**

   a) ARMA(0,3)
   b) ARMA(3,0)
   c) ARMA(0,2)
   d) ** ARMA(2,0)
   e) ARMA(1,1)

5. Suppose a time series appears to have a quadratic trend, $m_t = \beta_0 + \beta_1 t + \beta_2 t^2$. What transformations can be used to eliminate that trend?
   Choose all that are appropriate including NONE

   a) Apply a Box-Cox transformation with an appropriately chosen λ parameter.

b) Apply a moving-average smoother

c) ** Difference the series twice

6. Suppose that $\{X_t\}$ is a stationary ARMA(p,q) time series. Which of the following is/are true? s and t are integers.
Choose as many as are appropriate including NONE

 a) $E(X_t|\mathcal{F}_{t-1}) = E(X_s|\mathcal{F}_{s-1})$ for all s and t.

 b) ** $Var(X_t|\mathcal{F}_{t-1}) = Var(X_s|\mathcal{F}_{s-1})$ for all s and t.

 c) $Y_t = \log(X_t)$ also follows a stationary ARMA model.

 d) The theoretical PACF of the model is 0 at all lags larger than $p + q$.

7. The Dickey Fuller test was applied to the time series dataset **X**, and it resulted in a Dickey Fuller statistic of -6.81 with p-value 2.09e-09.

 What is the most appropriate conclusion to draw from this result?
Choose exactly one answer

 a) There is strong evidence that the series X is not stationary.

 b) **There is strong evidence that the series X is stationary.

 c) There was a failure to find strong evidence that the series is stationary.

 d) The series must be differenced at least once before it is stationary.

8. If a time series $\{X_t\}$ is known to be both stationary and conditionally heteroskedastic, this implies which of the following?
Choose as many as are appropriate including NONE

 a) $Var(X_t)$ is a non-constant function of t?

 b) **$Var(X_t|\mathcal{F}_{t-1})$ is a non-constant function of t.

9. What is the **leverage effect** when referring to the stylized facts for stock price behavior?
Choose exactly one answer

 a) There are sharp changes ("jumps") in the volatility of a price series.

 b) Volatility tends to increase more after a large price increase than it does after a large price decrease.

 c) ** Volatility tends to increase more after a large price decrease than it does after a large price increase.

 d) Volatility tends to increase with an increase in traded volume.

10. An MSCF student has downloaded data on a particular stock price from 1/1/2000 to 12/31/2019. In looking over these data, the student notices that some of these data appear to have unusually large values and the student changes some of the values and then goes on to perform certain statistical analyses. What statistical activity is the student engaged in by changing some of the data values?
Choose exactly one answer

a) Prediction

b) Filtering

c) ** Smoothing

d) Seasonally adjusting

11. Which of the following stylized facts is an ARMA+GARCH model intended to address?
Choose as many as are appropriate including NONE

    a) ** Heavy tailed distribution of log-returns

    b) Gains/Loss Asymmetry

    c) ** Volatiity Clustering

    d) Leverage Effect

12. Which of the following procedures could be used to determine if there are ARCH effects in a time series?
Choose all that are appropriate including NONE

    a) Use the adjusted Dickey Fuller test on the squared residuals

    b) ** Use the Engle test on the squared residuals

    c) ** Use the Ljung Box test on the squared residuals

    d) Use the Engle-Granger test on the squared residuals

13. Consider the following time series model presented in state space form, where $\{\epsilon_t\}$ is WN$(0, \sigma_\epsilon^2)$. $\mathbf{X}_t = (X_t, X_{t-1})^T, \mathbf{V}_t = (\epsilon_t + \theta_1 \epsilon_{t-1}, 0)^T, Y_t = (1,0)\mathbf{X}_t$, and $\mathbf{X}_t = \mathbf{F}\mathbf{X}_{t-1} + \mathbf{V}_t$, where

$$\mathbf{F} = \begin{bmatrix} \phi_1 & 0 \\ 1 & 0 \end{bmatrix},$$

What standard time series model does this correspond to?
(Choose exactly one answer

    a) AR(1)

    b) MA(1)

    c) ** ARMA(1,1)

    d) ARIMA(1,1,1)

14. Stock price data for 1/1/2010-12/31/2019 was downloaded for a collection of 10 stocks, all in the "tech" sector. The analyst who downloaded the data wants to know if any of the stocks are "moving together" in the sense that the value of a portfolio of some subset of them forms a stationary process. Such a finding would allow the analyst to construct a trading strategy or use time series methods to predict the future value of the portfolio. In order to determine whether there is/are any cointegrated subset(s) of stocks, the analyst will need to do, among other procedures:
Choose as many as are appropriate including NONE

a) Perform the Dickey Fuller test on each of the series to ensure they are individually stationary.

b) After fitting a time series model for the mean of each series, check each for ARCH effects using the Engle or Ljung Box test appropriately.

c) ** Perform the Dickey Fuller test on each series and difference them appropriately, then apply the Engle-Granger test to each pair of the collection of 10 stocks.

d) ** Perform the Dickey-Fuller test on each individual stock and difference them appropriately, then perform the Johansen test on the entire set of 10 stocks all at once.

15. Suppose a time series $\{X_t\}$ is being modeled using an ARMA(1,1)+ARCH(1). That is

$$X_t - \mu = \phi_1(X_{t-1} - \mu) + \theta_1 a_{t-1} + a_t,$$

where $a_t = \sigma_t \epsilon_t$, $\sigma_t = \sqrt{\omega + \alpha_1 a_{t-1}^2}$, and $\{\epsilon_t\}$ is WN$(0, \sigma_\epsilon^2)$ with $E(\epsilon_t^4) < \infty$.
Choose as many as are appropriate including NONE

a) $\{a_t\}$ is WN$(0, \frac{\phi_1}{1-\phi_1^2})$

b) ** $\rho_{a^2}(h) = \alpha_1^{|h|}$

c) ** $\{a_t^2\}$ can be represented as an AR(1) process

d) $Var(a_t)$ varies with t.

16. Suppose that $\{X_t\}$ is a time series model defined by

$$X_t = \mu + \epsilon_t - 2\epsilon_{t-1} + \epsilon_{t-2},$$

where $\{\epsilon_t\}$ is WN$(0, \sigma^2)$. In the space below, type in the autocovariance function of the X_t process at all lags.

$\gamma(0) = 6\sigma^2, \gamma(1) = -4\sigma^2, \gamma(2) = \sigma^2, \gamma(h) = 0$ if $|h| \geq 3$.

17. Suppose $\{X_t\}$ is an AR(1) time series,

$$X_t = \phi_0 + \phi_1 X_{t-1} + \epsilon_t,$$

where $|\phi_1| < 1$ and $\{\epsilon_t\}$ is WN$(0, \sigma_\epsilon^2)$.

In the space below, type in the PACF at all lags.

ϕ_1 at lag 1 and -1, 0 otherwise.

18. Suppose that $\{X_t\}$ is an ARMA(1,1) time series, specified by

$$X_t = \phi_0 + \phi_1 X_{t-1} + \theta_1 \epsilon_{t-1} + \epsilon_t.$$

where $\{\epsilon_t\}$ is WN$(0, \sigma_\epsilon^2)$. Suppose we have information on $\{X_t, \ 0 \leq t \leq T\}$, and estimates for the ARMA parameters based on those data. In the space below, write an expression for the minimum mean squared error estimate of X_{T+1}, i.e. $\hat{X}_{T+1}$

$$\hat{X}_{T+1} = \hat{\phi}_0 + \hat{\phi}_1 X_T + \hat{\theta}_1 \hat{\epsilon}_T.$$